

EDUCATION

Ph.D in Computer Science - Worcester Polytechnic Institute. Aug. 2019 - present
Research focus: *Data Visualization and Human-Computer Interaction* GPA: 4.0/4.0
Advisor: Prof. Lane Harrison

Masters in Computer Science - Worcester Polytechnic Institute. GPA: 4.0/4.0
Selected courses: Data visualization, Design & Analysis of large experiments, Algorithm design & analysis, Software design & analysis,

National Diploma of Telecommunications Engineering 2010 - 2015
Higher School of Communications of Tunis, Tunisia
Major: *Telecommunications and cybersecurity*
Thesis: Design and Evaluation of Redactable Signature algorithms on IoT devices [4]

Ph.D. RESEARCH PROJECTS

Natural language and data visualizations

I study the interplay of natural language with data visualization, and the challenges or opportunities that the combination of the two raise across different cultures and languages. Example projects:

- I study the perception of probabilistic language through charts and how they vary across languages [1]
- I study the impact of dual languages on the interpretation of data visualization [2]
- I collect eye-tracking data to investigate how people navigate and learn from interactive interfaces in languages with different writing & reading directions

Cross-cultural data visualization

Through a computational analysis of social media data, and interviews with visual data designers, I investigate the use of dashboard visualization across regions of the world, the characteristics of design and data rhetoric that can surface across cultures.

Data Visualization for cybersecurity

I interview security analysts to investigate the organizational and technical challenges and barriers that they face when adopting state-of-the-art visual analytics tools and best-practices [9]

EXPERTISE

Visualization and UI design: JavaScript, ReactJS, D3js, Java, Figma, Adobe Illustrator, AdobeXD

Experiment design: user study, qualitative & mixed method, cross-language study, eye-tracking study

Experiment design: statistical inference, Natural language interface, NLP, R, NVIVO, AWS

Software design and analysis: Java, Object-oriented programming

PROFESSIONAL EXPERIENCES

VISITING RESEARCHER

I conducted research at the Data Visualization and Analysis Group

University of Konstanz, Germany
Feb. - Aug. 2023

TEACHING ASSISTANTSHIP (TA)

I assisted the following classes:

Worcester Polytechnic Institute
Fall 2019 - Fall 2022

2019-2020: Web Development, Software Engineering, HCI, Data Visualization,

2020-2021: Operating Systems, Algorithms, Advanced Object-Oriented Programming, Data Visualization, Computer Networks

2021-2022: Foundations of Computer Science, Algorithms, Human-Computer Interaction, Data Visualization,

Fall 2022: Web Development, Software Engineering

Award: Best TA of the year 2020 (CS dept.), 2021 (Honorable Mention College-wide)

RESEARCH ASSOCIATE

University of Passau, Germany

I contributed the following projects:

2015 - 2019

FORSEC project (2015-2017): I developed a data visualization architecture for low-level security data in IaaS cloud [5] collected using Virtual Machine Introspection techniques [6]

DINGFEST project (2018-2019): I studied the design and evaluation of visual analytics components in next-generation SIEM systems [7]

ARADIA project (2018-2019): I designed a component-based dashboard with linked multiple views of multi-sources data to support cyber security monitoring tasks [8] within an enterprise network

OpenC3S (2017-2019): I co-designed syllabi and teaching materials for Cloud Computing Security classes as part of the “largest basic and advanced education initiative in German-speaking countries in the area of cyber security”

Tools & languages: C++, JavaScript, React.js, D3.js, Tableau, IaaS cloud (OpenNebula, AWS), Java

LANGUAGES

I speak **Malagasy, French, English, German, Portuguese (BR), and Arabic (TN)**

SERVICES and OUTREACH

- Madagascar R users group: Co-Founder, outreach and international cooperation chair
- ASSETS 2022: student volunteer
- IEEE VIS: student volunteer 2020 - 2022, incoming SV chair 2023
- IEEE Symposium on Visualization for Cyber Security: Organizing Committee 2019-2022
- Data Visualization Society (DVS): editor for Nightingale, organizing committee for Outlier Conference
- Ikala STEM (Madagascar-based non-profit association for empowering female Malagasy Scientists): managing editor for Ikala STEM's yearly journal, mentor at Ikala STEM mentoring program, US resident member (prev. EU resident member)
- First series of IT-security summer school in Madagascar: Project leader of the 2018-2020 editions
- AnitaB.org mentorship program: Mentor

AWARDS

- WPI Diversity, Equity, Inclusion and Justice planning grant 2022 (\$4000)
- WPI teaching assistant of the year: 2020, 2021 (honorable mention college-wide)
- Women in Cyber Security WiCyS scholar (Aurora CO, 2020)
- Grace Hopper Celebration scholarship (Houston TX, Sep. 2018)
- Young Security Changers Scholarship at the Munich Security Conference (Germany, 2018)
- Diversity scholarship VizSec (Oct. 2017) at the Symposium on Visualization for Cyber Security (Phoenix AZ, Sept. 2017)
- DAAD exchange student scholarship, (Passau Germany, May-Sep 2015)

SELECTED PUBLICATIONS & POSTERS

Complete list of publication available on [DBLP](#) and [Scholar](#)

- [1] [N. Rakotondravony](#), Y. Ding, L. Harrison: “Probablement, Wahrscheinlich, Likely? A Cross-Language Study of How People Verbalize Probabilities in Icon Array Visualizations”. IEEE Transactions on Visualization and Computer Graphics (TVCG) 2022
- [2] H. Andrianarivony, T. Raharison, I. Rakotoniaina, M. Ramarokoto, [N. Rakotondravony](#), L. Harrison: “Investigating the Use of Native and Secondary Language with Data Visualization in Madagascar”. IEEE VIS 2022 - poster track
- [3] R. Birchfield, M. Caten, E. Cheng, M. Kelly, T. Larson, H. P. Pham, Y. Ding, [N. Rakotondravony](#), L. Harrison: “VisQuiz: Exploring Feedback Mechanisms to Improve Graphical Perception”. IEEE Transactions on Visualization and Computer Graphics (TVCG) 2022
- [4] C. Frädrih, H. C. Pöhls, W. Popp, [N. Rakotondravony](#), Kai Samelin.: “Integrity and authenticity protection with selective disclosure control in the cloud & IoT”. In International Conference on Information and Communications Security. Springer, Cham, 2016.
- [4] [N. Rakotondravony](#), Lane Harrison. “Visualization for Cyber Security and Security Analysts’ Use of Visualizations: Is There a Gap?” Poster at IEEE Symposium on Visualization for Cyber Security, 2020.
- [5] [N. Rakotondravony](#) and H. P. Reiser, “Visualizing and Controlling VMI-based malware analysis in IaaS Cloud.” In Symposium on Reliable Distributed Systems (SRDS), 2016.
- [6] B. Taubmann, [N. Rakotondravony](#) and H. P. Reiser, “CloudPhylactor: Harnessing Mandatory Access Control for Virtual Machine Introspection in Cloud Data Centers”. In the 15th IEEE International Conf. on Trust, Security and Privacy in Computing and Communications, 2016.
- [7] F. Böhm, [N. Rakotondravony](#), G. Pernul, H. P. Reiser, “Exploring the roles of experts’ knowledge in cyber security visualizations.” Poster at the IEEE Symposium on Visualization for Cyber Security, 2018.
- [8] [N. Rakotondravony](#) and H. P. Reiser, “Visualizing BFT SMR distributed systems - example of BFT-SMaRt”. In DSN Workshop on Byzantine Consensus and Resilient Blockchains, 2018.
- [9] M.A. Puentes, Y. Lei, [N. Rakotondravony](#), L.T. Harrison, C.A. Shue, “Visualizing Web Application Execution Logs to Improve Software Security Defect Localization”. 5th Workshop on Validation, Analysis and Evolution of Software Tests, 2022.